
Math Training — M1 Cognitive Science

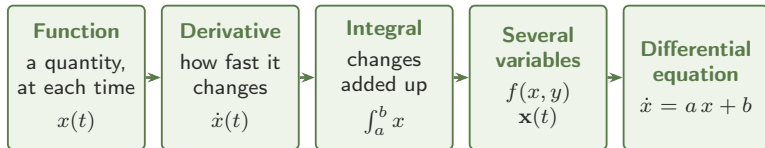
Session 5 — Analysis

Functions, derivatives, integrals, function of several variables, differential equation

Basic track

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Roadmap: from functions to differential equations



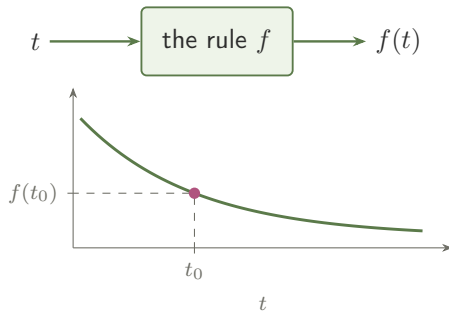
Each box uses the one before it. The last box is the destination: an equation whose unknown is not a number but a whole *function*.

■ Definition

A **function** f is a rule that turns one number into one number: give it t , it returns $f(t)$. The **graph** of f is the set of points $(t, f(t))$: above each t , a point at height $f(t)$.

In $f(t) = 0.9e^{-t/2}$ there is one **variable**, t , and two **constants**, 0.9 and 2.

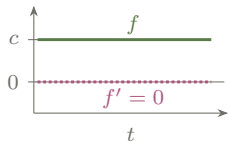
Changing t moves you *along* the curve. Changing a constant gives a *different* curve — and fitting a model to data means choosing the constants.



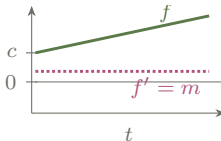
Graph of $t \mapsto 0.9e^{-t/2}$. Read it in two moves: along to t_0 , then up to the curve. The height reached is $f(t_0)$, and the pink point $(t_0, f(t_0))$ is one point of the graph.

Three functions to recognise, and their derivatives

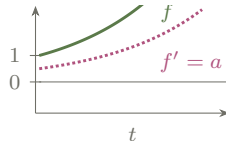
Constant: $f(t) = c$



Affine: $f(t) = mt + c$



Exponential: $f(t) = e^{at}$



In green the function, in **pink dotted** its derivative, on the same axes. **Left:** a constant does not change, so its derivative is the zero function. **Centre:** an affine function climbs at a fixed rate, so its derivative is the constant m . **Right:** the derivative of e^{at} is $a e^{at}$ — the *same shape*, scaled by a (here $a = \frac{1}{2}$, so the dotted curve is the green one halved). In the first two panels the derivative is a horizontal line; in the third it is a curve.

The exponential is the odd one out, and that is exactly why it matters: its derivative is **proportional to the function itself**. Keep that sentence — the last part of the session is built on it.

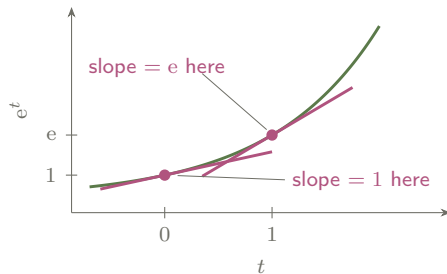
The exponential: one property, three consequences

■ Definition

e^t is the function **equal to its own derivative**: at every point, its slope equals its height. The number $e \approx 2.718$ is its value at $t = 1$.

Three consequences, and they are all we use:

- $e^0 = 1$;
- $e^{at} > 0$ for every t : it **never reaches zero**;
- the **sign of a** decides the direction (growing if $a > 0$, decaying if $a < 0$); its *size* decides only how fast.



The defining property, drawn twice. Where the curve has height 1, the tangent has slope 1; where it has height e , the tangent has slope e . Height and slope are the same number, everywhere on the curve.

The decaying exponential and its time constant τ

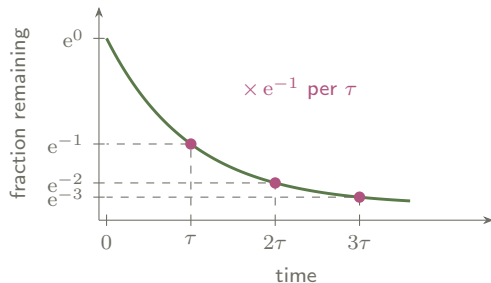
■ Definition

When $a < 0$, write $a = -1/\tau$, so that the decay reads $e^{-t/\tau}$. The positive number τ is the **time constant**, measured in the same units as t .

■ Intuition

Waiting one τ always multiplies what is left by $e^{-1} \approx 0.37$, whenever you start counting.

So a **large τ means slow decay** — the opposite of what most people expect the first time.



Graph of $t \mapsto e^{-t/\tau}$. Each step of length τ multiplies the height by e^{-1} , so the heights are $e^0, e^{-1}, e^{-2}, e^{-3}$, that is about 100%, 37%, 14%, 5%. The curve falls towards zero and never reaches it.

■ Link to cognitive science

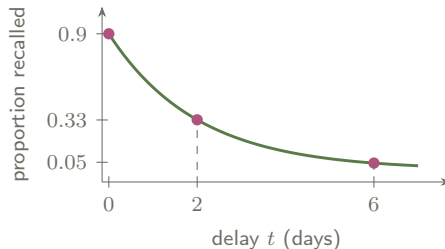
Learn a list of words, then test recall after a delay t . The proportion still recalled decreases as

$$R(t) = R_0 e^{-t/\tau}.$$

Two constants, and the whole curve is fixed:

- R_0 : what is recalled immediately, at $t = 0$;
- τ : how quickly recall falls.

Fitting this model to data means choosing those two numbers.



The case $R_0 = 0.9$ and $\tau = 2$ days. The three pink points are $t = 0$, $t = \tau$ and $t = 3\tau$: one time constant leaves 37% of R_0 , three leave about 5%.

■ Practice — try it

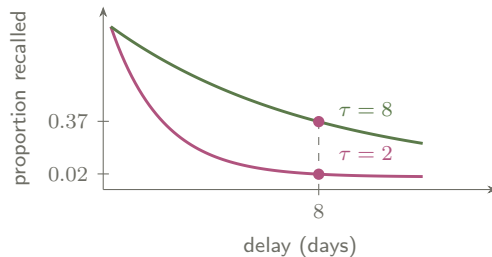
1. Does e^{3t} increase or decrease? And e^{-3t} ?
2. Write $e^{-t/5}$ in the form e^{at} , and give a and τ .
3. Recall follows $R(t) = 0.8e^{-t/3}$, with t in days. Give $R(0)$, $R(3)$ and $R(9)$.
4. Two students forget with $\tau = 2$ days and $\tau = 8$ days. After 8 days, who recalls more?

■ What these exercises use

- The **sign of a** decides increasing or decreasing; its size decides only how fast.
- $e^{-t/\tau}$ is e^{at} with $a = -1/\tau$, so $\tau = -1/a$.
- After n time constants, what remains is R_0 multiplied by e^{-n} .

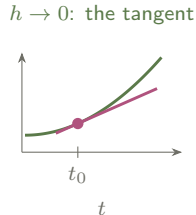
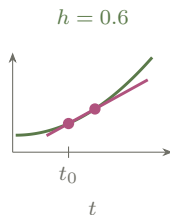
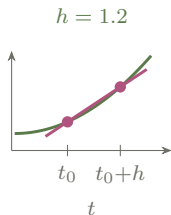
1. e^{3t} increases, e^{-3t} decreases.
2. $a = -1/5$ and $\tau = 5$.
3. $R(0) = 0.8$, $R(3) \approx 0.30$,
 $R(9) \approx 0.04$.
4. The student with $\tau = 8$.

A **larger** τ means **slower** forgetting.



Question 4. At $t = 8$ days the green curve has run through *one* time constant and keeps 37%; the pink curve has run through *four* and keeps about 2%.

Definition of the derivative: from average to instantaneous rate



The same curve three times. The pink line joins the point at t_0 to the point at $t_0 + h$; its slope is the *average* rate of change over that interval. As h shrinks, the second point slides back and the line settles on one limiting position: the tangent at t_0 .

■ Definition

The **derivative** of f at t_0 is the slope of that limiting line:

$$f'(t_0) = \lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h} \quad (\text{rise divided by run, as the run shrinks}).$$

Notation and units of the derivative

Three notations, one object:

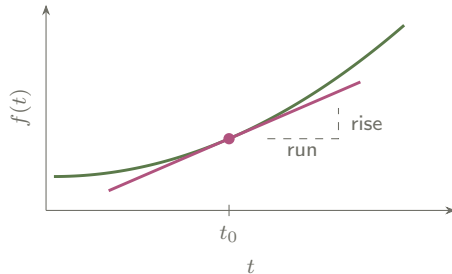
$$f'(t_0), \quad \frac{df}{dt}(t_0), \quad \dot{f}(t_0).$$

The **dot** is used when the variable is time; it is the notation of the last part of today.

■ Intuition

A derivative is a **rate**, and it carries units: if f is a number of words and t is in days, then f' is in words per day.

So f' answers: *if I wait a little, by how much does f change, per unit of waiting?*



The pink line is the **tangent** to the green curve at $(t_0, f(t_0))$: it touches the curve there and has the same direction. Its slope is rise divided by run, and that number is $f'(t_0)$.

The derivative as a function: sign, growth and extrema

Every point of the curve has its own slope. Collecting all of them gives a **new function**, f' , with its own graph.

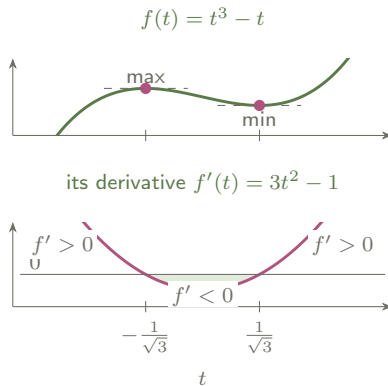
■ The one rule to remember

$f' > 0 \Rightarrow f$ goes up,

$f' < 0 \Rightarrow f$ goes down,

$f' = 0 \Rightarrow f$ is flat there.

To find a maximum or a minimum: solve $f' = 0$, then read the **sign of f'** just before and just after.



Same horizontal axis, read together. Where the pink curve is above zero the green curve climbs; where it is below, the green curve falls; where it crosses zero, the green curve turns.

■ Useful fact

$$\text{a constant} \rightarrow 0, \quad t^n \rightarrow n t^{n-1}, \quad \boxed{e^{at} \rightarrow a e^{at}}$$

A sum is differentiated **one term at a time**, and a constant factor is kept: $(3f)' = 3f'$.
The power rule also holds for negative and fractional n , so $1/t \rightarrow -1/t^2$ and $\sqrt{t} \rightarrow 1/(2\sqrt{t})$.

■ Intuition

The boxed rule in words: differentiating an exponential returns the *same* exponential, multiplied by the number sitting in front of t in the exponent.

Two checks: $e^t \rightarrow e^t$ (here $a = 1$: the defining property again), and $e^{-t/\tau} \rightarrow -\frac{1}{\tau} e^{-t/\tau}$ (the forgetting curve).

Apart from the constant function 0, exponentials are the **only** functions whose derivative is proportional to the function itself. That is why they will be the answer at the end of the session.

A frequent mistake: power rule versus exponential rule

■ Watch out

$e^{-t/2}$ is **not a power of t** , so the power rule does not apply to it. Its derivative is $-\frac{1}{2}e^{-t/2}$, and *not* $(-t/2)e^{(-t/2)-1}$.

Before differentiating, one question is enough:

where does t stand?

t in the **base** $\Rightarrow$ power rule

$$t^2 \rightarrow 2t$$

t in the **exponent** $\Rightarrow$ boxed rule

$$e^{2t} \rightarrow 2e^{2t}$$

The position of the variable decides the rule. In t^2 the variable is the thing being squared; in e^{2t} the variable is up in the exponent, and the base is the fixed number e .

■ Example

Differentiate $f(t) = 3t^2 + 2t + 5$, $R(t) = 0.9e^{-t/2}$, $P(t) = 100 - 80e^{-t/10}$.

A polynomial, one term at a time: $3t^2 \rightarrow 6t$, $2t \rightarrow 2$, $5 \rightarrow 0$, hence $f'(t) = 6t + 2$.

An exponential: the number in front of t in the exponent is $a = -\frac{1}{2}$, hence

$$R'(t) = -\frac{1}{2} \cdot 0.9e^{-t/2} = -\frac{1}{2} R(t).$$

In words: the speed at which recall falls is proportional to what is still recalled. Keep this line — it is already a **differential equation**.

A constant plus an exponential: the constant gives 0, and the two minus signs combine, hence $P'(t) = +8e^{-t/10}$: positive for every t , so P always increases, and largest at $t = 0$, so it increases fastest at the start.

■ Practice — try it

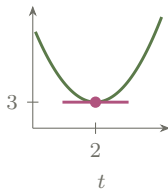
1. Differentiate $f(t) = t^2 - 4t + 7$. Where is its slope zero, and is that a maximum or a minimum?
2. Differentiate $g(t) = e^{3t}$ and $h(t) = 5e^{-2t}$. Which one increases?
3. Differentiate $R(t) = 0.8e^{-t/4}$ and check that $R' = -R/4$.
4. Differentiate $P(t) = 100 - 60e^{-t/5}$. Is P increasing or decreasing?

■ What these exercises use

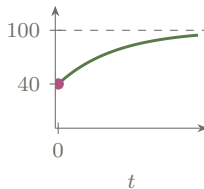
- **One term at a time**, then solve $f' = 0$ and read the sign of f' on each side.
- **The boxed rule**, $ce^{at} \rightarrow ace^{at}$: the constant c and the exponent are unchanged, and a comes down in front. Watch the case where c and a are **both negative**.

1. $f' = 2t - 4$, zero at $t = 2$;
negative before and positive
after, so a **minimum**.
2. $g' = 3e^{3t}$: increases.
 $h' = -10e^{-2t} < 0$: decreases.
3. $R' = -\frac{1}{4} \cdot 0.8e^{-t/4} = -\frac{1}{4}R$.
4. $P' = +12e^{-t/5} > 0$: increasing.

question 1



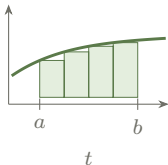
question 4



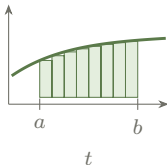
Question 1: the tangent at $t = 2$ is horizontal and the curve lies above it on both sides — that is what a minimum is. Question 4: P starts at 40 and rises towards the dashed ceiling 100, so its derivative is positive everywhere.

Definition of the integral: a limit of sums of slices

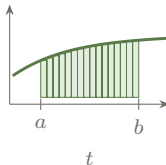
4 rectangles



8 rectangles



16 rectangles



Cut $[a, b]$ into slices. Each slice is almost a rectangle: width dt , height $f(t)$, area $f(t) \times dt$. Adding them approximates the area under the curve, and the thinner the slices, the better the approximation.

■ Definition

$\int_a^b f(t) dt$ is the **limit of these sums** as the slices become infinitely thin. It is the **area** between the graph of f and the horizontal axis, from $t = a$ to $t = b$. The sign $\int$ is a stretched S, for *sum*.

The integral is a signed area

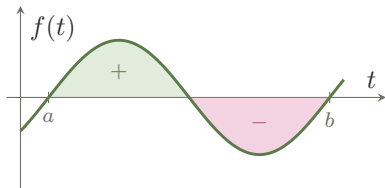
■ Definition

In $\int_a^b f(t) dt$, the part of the region **above** the axis counts positively and the part **below** counts negatively. The integral is the *net* area.

The reason is in the slices: a slice has area $f(t) \times dt$, and when $f(t)$ is negative that product is negative.

■ Intuition

If f is a **rate** — words learnt per day, current flowing in — then the integral is the **total accumulated**, and a negative rate removes from the total.



Here the two regions have the same shape, one above the axis and one below, so they cancel: $\int_a^b f = 0$. The function was certainly not zero — the integral only says that what was gained was later lost.

The running total and the fundamental theorem

Fix the left end a and let the right end move. This defines a new function,

$$F(t) = \int_a^t f(s) ds,$$

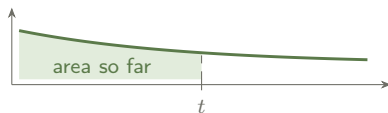
the area **accumulated so far**.

■ Fundamental theorem

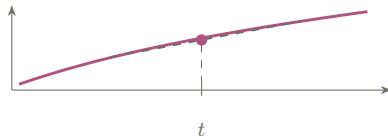
$F' = f$: the rate at which the accumulated area grows is exactly the height of the curve.

In one line: moving the right end by dt adds a sliver of area $f(t) \times dt$, so $dF/dt = f(t)$.

the rate $f(t)$



the running total $F(t)$



Top: the area collected up to t . Bottom: that area, plotted against t ; the dashed line is its tangent at the same t . The bottom curve climbs steeply while the top curve is high and flattens as the top curve comes down — because its slope *is* the height above.

■ Useful fact

An **antiderivative** of f is any function F with $F' = f$. Two antiderivatives of the same f differ only by a constant, which is why one writes $\int f(t) dt = F(t) + C$. For a definite integral,

$$\int_a^b f(t) dt = F(b) - F(a),$$

and the constant **cancels** in the subtraction, so *any* antiderivative may be used.

The table of derivatives, read from right to left, is a table of integrals:

$$\int t^n dt = \frac{t^{n+1}}{n+1} \quad (n \neq -1), \quad \int e^{at} dt = \frac{e^{at}}{a}, \quad \int \frac{dt}{t} = \ln |t|.$$

■ Three steps, in this order

(i) find F with $F' = f$; (ii) evaluate F at b ; (iii) subtract $F(a)$. Then **check by differentiating F** : you should get f back.

■ Example

Compute $\int_0^2 (3t^2 - 4t + 1) dt$ and $\int_0^T e^{-t/\tau} dt$.

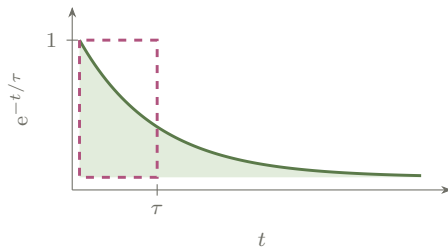
First: an antiderivative is

$F(t) = t^3 - 2t^2 + t$, so the integral is $F(2) - F(0) = (8 - 8 + 2) - 0 = 2$.

Second: an antiderivative of $e^{-t/\tau}$ is $-\tau e^{-t/\tau}$ (divide by $a = -1/\tau$), so

$$\int_0^T e^{-t/\tau} dt = \tau(1 - e^{-T/\tau}),$$

which grows towards τ as T increases.



The green region continues to the right without end, yet its total area is finite: exactly the area of the dashed rectangle, height 1 and width τ . So the time constant is also an area.

■ Practice — try it

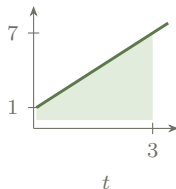
1. Compute $\int_0^3 (2t + 1) dt$.
2. Compute $\int_0^\infty e^{-2t} dt$. What does the answer equal, in words?
3. Find an antiderivative of $e^{-t/5}$, and check it by differentiating.
4. A rate is $f(t) = 0.35$ for $0 \leq t \leq 4$. Without any calculation, what is $\int_0^4 f$? What shape is the region?

■ What these exercises use

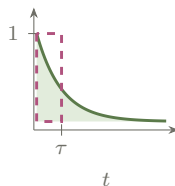
- **The three steps.** In 1: $F(t) = t^2 + t$, so $F(3) - F(0) = (9 + 3) - 0 = 12$.
- **The exponential integral** $\int e^{at} dt = e^{at}/a$: divide by a , keeping its sign. In 2, $e^{-2t} \rightarrow 0$ as t grows, so only the lower limit contributes.
- An area can sometimes just be **read off**, as in 4.

1. $F(t) = t^2 + t$, and $F(3) - F(0) = 12$.
2. $\frac{1}{2}$ — the time constant of e^{-2t} .
3. $-5e^{-t/5}$, whose derivative is $e^{-t/5}$. ✓
4. $0.35 \times 4 = 1.4$; the region is a rectangle.

question 1



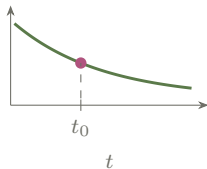
question 2



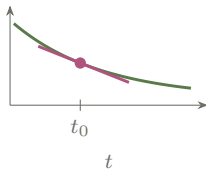
Question 1: the region is a trapezium of width 3 whose parallel sides have lengths 1 and 7, so its area is $3 \times (1 + 7)/2 = 12$. Question 2: the same picture as the worked example, with $\tau = \frac{1}{2}$.

Recap: function, derivative and integral on one curve

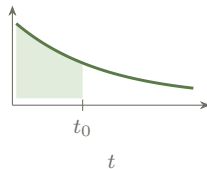
the function $x(t)$



its derivative at t_0



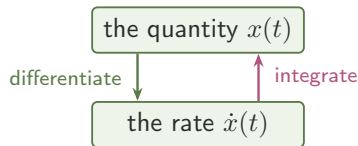
its integral from 0 to t_0



One and the same function, asked three questions. **Left:** what is its *value* here? **Centre:** how *steep* is it here — the slope of the pink tangent. **Right:** how much has *accumulated* up to here — the green area.

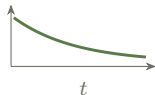
■ Differentiating and integrating undo each other

Differentiate x , then integrate: you are back to x , up to **one constant**. That constant is the value of x at the starting time — a rate alone cannot know where the quantity started.



Two ways of having more than one number

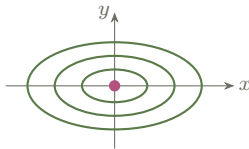
Until now: one number in, one number out. Two different things can change.



■ One in, one out

$f(t)$, a **curve**.

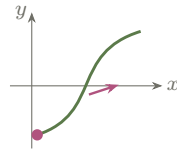
Derivative: one number per point, $f'(t)$.



■ Several in, one out

$f(x, y)$, a **surface**, drawn as a contour map.

Derivative: **one per input** — the partial derivatives.



■ One in, several out

$\mathbf{x}(t)$, a **path** in the plane.

Derivative: **one per output** — the velocity vector.

■ Key idea

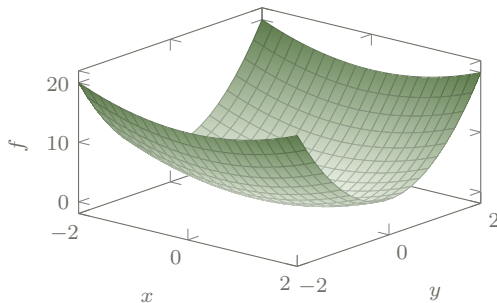
There is **no new calculus** in this part: every derivative below is the one-variable derivative of the last hour, applied one letter at a time. What is new is only how the results are *collected*.

■ Definition

A function of **several variables** takes several numbers and returns one number, written $f(x, y)$. Its graph is a **surface**: above each point (x, y) of the plane, a height $z = f(x, y)$.

This is the normal situation in an experiment: accuracy depends on difficulty *and* on practice; the error of a model depends on all of its parameters at once.

Beyond two inputs the surface cannot be drawn — and nothing that follows needs the drawing.



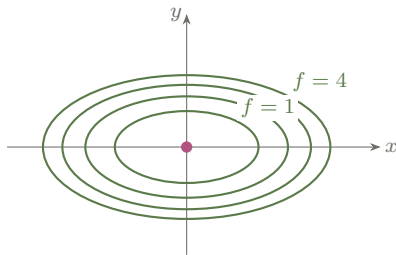
The graph of $f(x, y) = x^2 + 4y^2$ is a bowl, with its lowest point at the origin. The same f is used on the next two frames, so that the surface, its contour map and its partial derivatives are three views of *one* object.

■ Definition

A **contour line** is the set of points where f takes one fixed value: all (x, y) with $f(x, y) = c$. One line per value of a chosen list gives the **contour map**.

Read it exactly like a walking map, with altitude in place of f :

- lines **close together**: steep;
- lines **far apart**: flat;
- loops shrinking to a point: a summit or a hollow.



Contour map of the same $f(x, y) = x^2 + 4y^2$, for $f = 1, 2, 3, 4$; the pink dot at the centre is $f = 0$. Walking *along* one ring does not change f at all; f changes only when you step from one ring to the next. The rings are closer together vertically, so the bowl is steeper in the y direction.

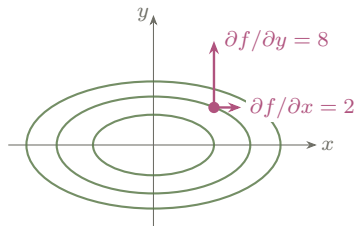
■ Definition

$\frac{\partial f}{\partial x}$ is the derivative of f with respect to x , every other variable being **held fixed**: differentiate with the ordinary rules, treating the other letters as numbers.

■ Example

For $f(x, y) = x^2 + 4y^2$: $\frac{\partial f}{\partial x} = 2x$
and $\frac{\partial f}{\partial y} = 8y$.

At $(1, 1)$ they are 2 and 8: the bowl is **four times steeper** in y , which is what the contour map already showed.

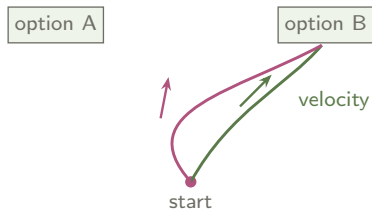


At the marked point, the two partial derivatives measure the steepness along the two arrows, whose lengths are proportional to those two numbers.

■ Definition

A **vector-valued function** returns several numbers at once, $\mathbf{x}(t) = (x(t), y(t))$. It is differentiated **one component at a time**: $\dot{\mathbf{x}}(t) = (\dot{x}(t), \dot{y}(t))$, which is again a vector.

As t increases, $\mathbf{x}(t)$ draws a **path**. The vector $\dot{\mathbf{x}}(t)$ is the **velocity**: its direction is the direction of travel, and its length, $\sqrt{\dot{x}^2 + \dot{y}^2}$, is the speed.



Two recorded cursor paths, both ending on option B; the arrows are velocity vectors, tangent to the path. The pink path first moves towards option A, which is read as competition between the two responses.

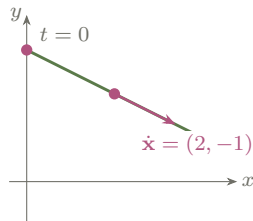
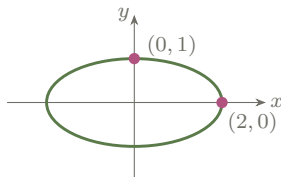
■ Practice — try it

1. For $f(x, y) = x^2 + 3xy + y$, compute both partial derivatives and their values at $(1, 2)$.
2. For $f(x, y) = x^2 + 4y^2$, write the equation of the contour line $f = 4$ and say what curve it is.
3. For $\mathbf{x}(t) = (2t, 3 - t)$, compute $\dot{\mathbf{x}}$ and the speed. What shape is the path?
4. For $\mathbf{x}(t) = (t, e^{-t})$, compute $\dot{\mathbf{x}}$ and give the velocity at $t = 0$.

■ What these exercises use

- Treat the other letter as a **number**, differentiate, and substitute the values of the point only at the very end. In $3xy$, the y is a **multiplier**: $\partial/\partial x$ gives $3y$, not 3.
- A contour line is $f(x, y) = c$; dividing that equation by c usually makes the curve recognisable.
- A vector-valued function is differentiated **component by component**; the speed is $\sqrt{\dot{x}^2 + \dot{y}^2}$, the length of a vector from Session 1.

1. $(2x + 3y, 3x + 1)$, equal to $(8, 4)$ at $(1, 2)$.
2. $x^2 + 4y^2 = 4$, that is $\frac{x^2}{4} + y^2 = 1$: an **ellipse** through $(\pm 2, 0)$ and $(0, \pm 1)$.
3. $\dot{\mathbf{x}} = (2, -1)$ for every t ; speed $\sqrt{5}$; the path is a **straight line**.
4. $\dot{\mathbf{x}} = (1, -e^{-t})$, equal to $(1, -1)$ at $t = 0$.



Question 2: the contour $f = 4$ is an ellipse, twice as wide as it is tall. Question 3: the path is a straight line, and the velocity is the *same* vector at every point of it.

*We are told how fast a quantity is changing, in terms of the quantity itself.
We want to know what the quantity does over time.*

■ Definition

A **differential equation** is an equation whose unknown is a **function**. It relates the function to its own derivative, as in

$$\dot{x}(t) = a x(t) + b.$$

To **solve** it is to find a function $x(t)$ whose derivative really is, at every time t , what the right-hand side prescribes.

■ Watch out

The unknown is a **function of t** , not a number: the answer looks like $x(t) = 3 - 3e^{-t}$, never like a single value.

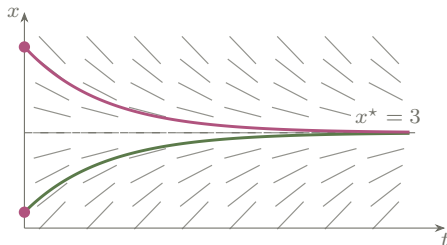
And do not integrate the right-hand side. Antidifferentiating solves $\dot{x} = f(t)$, where the rate depends on *time*; here the rate depends on x itself.

Reading the equation as a picture: the slope field

The equation gives a **slope at every point** of the plane: a solution through (t, x) must have slope $\dot{x} = ax + b$ there.

Draw a short segment of that slope at many points, and the solutions become visible before any formula is written: a **solution** is a curve that follows the segments.

Here $\dot{x} = -(x - 3)$. The slope depends only on the height x , so each horizontal row of segments is identical.



Slope field of $\dot{x} = -(x - 3)$, with two solutions drawn on it. Above the dashed line the segments point down, below it they point up, and on it they are horizontal — so every solution is pulled towards $x^* = 3$, from either side.

■ Definition

$$\frac{dx}{dt} = a x \implies x(t) = x_0 e^{at},$$

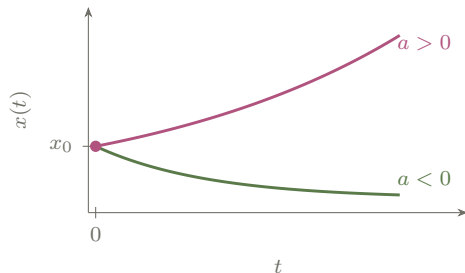
where $x_0 = x(0)$ is the value at the initial time.

Check it with the boxed rule:

differentiating $x_0 e^{at}$ gives $a x_0 e^{at}$, which is exactly $a x(t)$. ✓

The equation asks for a function whose derivative is a fixed multiple of itself, and **only exponentials do that**.

One such equation already appeared today:
 $R' = -R/\tau$.



Two solutions of $\dot{x} = ax$ starting from the same x_0 .
Only the sign of a differs: green decays towards zero, pink grows without bound.

The general case: $\dot{x} = ax + b$ and its steady state

■ Definition

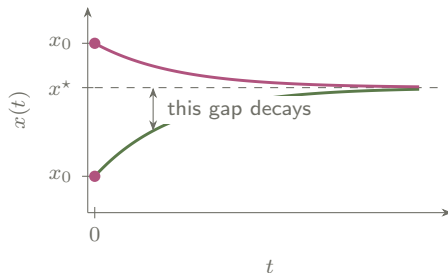
For $a \neq 0$,

$$\frac{dx}{dt} = ax + b \implies x(t) = x^* + (x_0 - x^*)e^{at},$$

with the **steady state** $x^* = -b/a$, the value at which the rate of change is zero.

Read the formula as a sentence: *the distance to x^* is multiplied by e^{at}* . Hence

- $a < 0$: the distance shrinks, x tends to x^* — **stable**;
- $a > 0$: the distance grows, x runs away — **unstable**.



The stable case $a < 0$, from two starting values. What decays exponentially is the *gap* between x and the dashed line x^* , from above and from below. x itself does not tend to zero.

Stability: the phase line

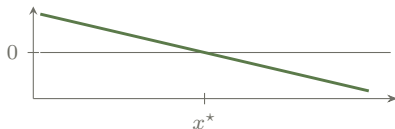
Plot the **rate** $\dot{x} = ax + b$ against x — not against t . It is a straight line, and it crosses zero exactly at x^* .

■ Intuition

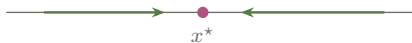
Where the line is above zero, $\dot{x} > 0$, so x moves to the **right**. Where it is below, x moves to the **left**.

Arrows pointing **towards** x^* : stable. Arrows pointing **away**: unstable. That is the sign of a , seen as a picture.

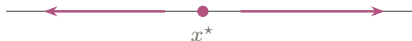
the rate $\dot{x}$ as a function of the state x



stable ($a < 0$): both arrows point in



unstable ($a > 0$): both point out



Top: the rate as a function of the state. Below: the same information as arrows on the x axis. A falling line gives inward arrows; a rising line gives outward ones.

■ Link to cognitive science

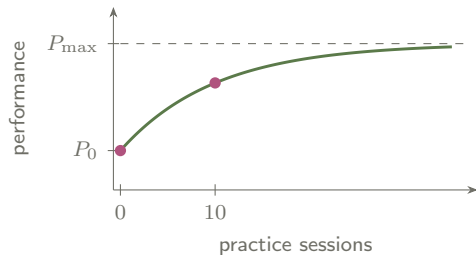
Performance improves with practice, but not without limit. There is a ceiling $P_{\max}$, and the closer performance gets to it, the slower the improvement:

$$\frac{dP}{dt} = k(P_{\max} - P), \quad k > 0.$$

This is $\dot{x} = ax + b$ with $a = -k$ and $b = kP_{\max}$, so $P^* = P_{\max}$, $\tau = 1/k$, and

$$P(t) = P_{\max} - (P_{\max} - P_0)e^{-kt}.$$

Forgetting and learning are the *same* equation read in two directions: in both, the rate is proportional to the distance that remains.



Performance rising towards its ceiling, with $\tau = 1/k = 10$ sessions. After one τ the remaining gap to $P_{\max}$ has shrunk to 37% of the initial gap.

■ With initial value $x(0) = x_0$

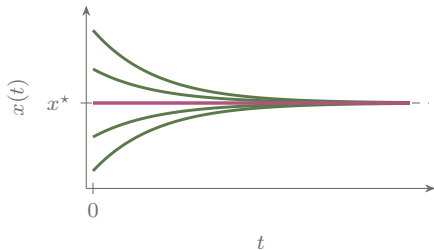
1. **Put the equation in the form** $\dot{x} = ax + b$, and read off a and b . (For $\dot{P} = k(P_{\max} - P)$: expand first, $a = -k$ and $b = kP_{\max}$.)
2. **Steady state:** $x^* = -b/a$ — equivalently, solve $\dot{x} = 0$.
3. **Time constant and stability:** $\tau = -1/a$; stable if $a < 0$, unstable if $a > 0$.
4. **Write the solution:** $x(t) = x^* + (x_0 - x^*)e^{at}$.

■ Two checks that cost nothing

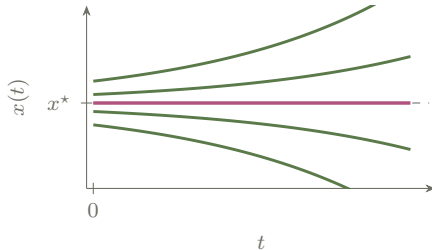
Put $t = 0$ in your answer: you must get x_0 back. Then differentiate your answer and compare with $ax + b$: they must agree for every t .

One equation, a family of solutions

$$\dot{x} = -(x - 3): \text{ stable}$$



$$\dot{x} = +(x - 3): \text{ unstable}$$



The same equation solved from several starting values. The initial value x_0 chooses *which* curve you are on; the coefficient a decides what the whole family does. **Left** ($a < 0$): every curve is drawn towards the pink steady state. **Right** ($a > 0$): every curve except the steady state itself is pushed away from it.

■ Practice — try it

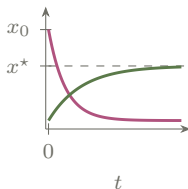
1. Solve $\dot{x} = -2x$ with $x_0 = 5$. What happens to x as t increases?
2. For $\dot{x} = -x + 3$: find x^* and τ , say whether it is stable, then write $x(t)$ starting from $x_0 = 0$.
3. Check by substitution that $x(t) = 3 + C e^{-t}$ solves $\dot{x} = -(x - 3)$ for every constant C .
4. Learning follows $\dot{P} = 0.2(90 - P)$ with $P_0 = 30$. Give τ and write $P(t)$.

■ What these exercises use

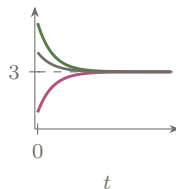
- **The four-step method:** form $\dot{x} = ax + b$; $x^* = -b/a$; $\tau = -1/a$ and the sign of a ; then $x(t) = x^* + (x_0 - x^*)e^{at}$.
- **To check a proposed solution:** differentiate it, compute the right-hand side separately, and compare the two.

1. $x(t) = 5e^{-2t}$, which tends to 0 (here $b = 0$, so $x^* = 0$).
2. $x^* = 3$, $\tau = 1$, **stable**;
 $x(t) = 3 - 3e^{-t}$.
3. $\dot{x} = -Ce^{-t}$, and
 $-(x - 3) = -Ce^{-t}$: equal for every C . ✓
4. $\tau = 1/0.2 = 5$ and
 $P(t) = 90 - 60e^{-t/5}$.

questions 1 and 2

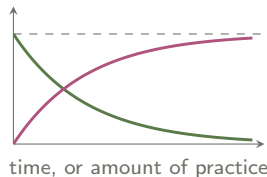


question 3



Questions 1 and 2: the pink curve tends to 0, the green one to $x^* = 3$ — same shape, different steady state.
Question 3: three members of the family $3 + Ce^{-t}$. All of them solve the equation; only the initial value fixes C .

- A **function** has one variable and some constants. The exponential e^{at} is the shape that matters, and its time constant $\tau = -1/a$ sets the scale.
- The **derivative** is the slope of the tangent, and a rate of change; its **sign** says increasing, decreasing or flat. Everything rests on one rule: $e^{at} \rightarrow a e^{at}$.
- The **integral** is a signed area, and an accumulated total. Compute it with an antiderivative: $F(b) - F(a)$.
- **Several inputs**: contour maps and partial derivatives, one letter at a time. **Several outputs**: a path and its velocity.
- $\dot{x} = ax + b$ has steady state $x^* = -b/a$, and the distance to x^* is multiplied by e^{at} — so x tends to x^* exactly when $a < 0$.



One differential equation, read in two directions: in **green**, forgetting; in **pink**, learning towards the dashed ceiling. Both obey “the rate is proportional to the distance that remains”, with the same τ .